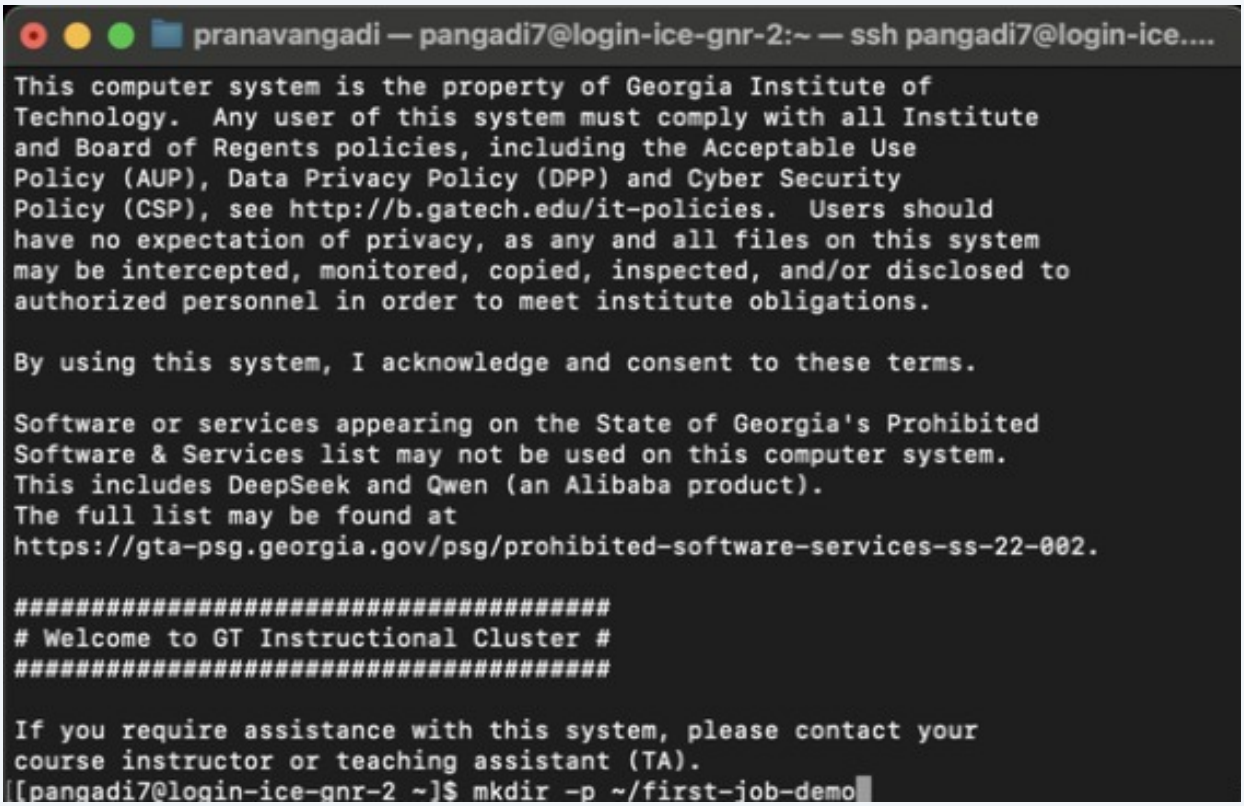
205 - Submitting Your First Job

Create a simple Slurm script, submit it with sbatch, monitor it with squeue, and read the resulting output file.

Directory Setup

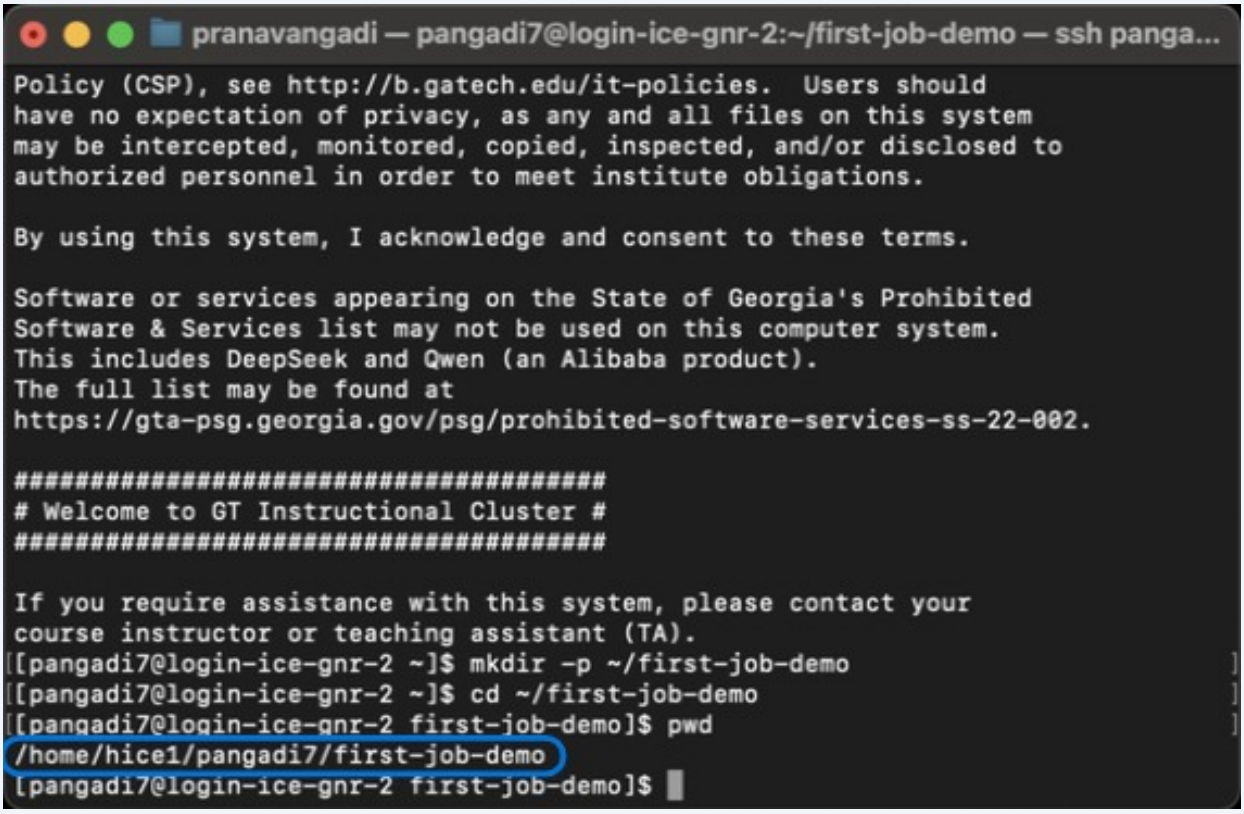
1 Create a folder for your first job
Type "**mkdir -p ~/first-job-demo Return**"



2 Open the first-job folder
Type "**cd ~/first-job-demo Return**"



3 Confirm your working directory
Type "**pwd Return**"

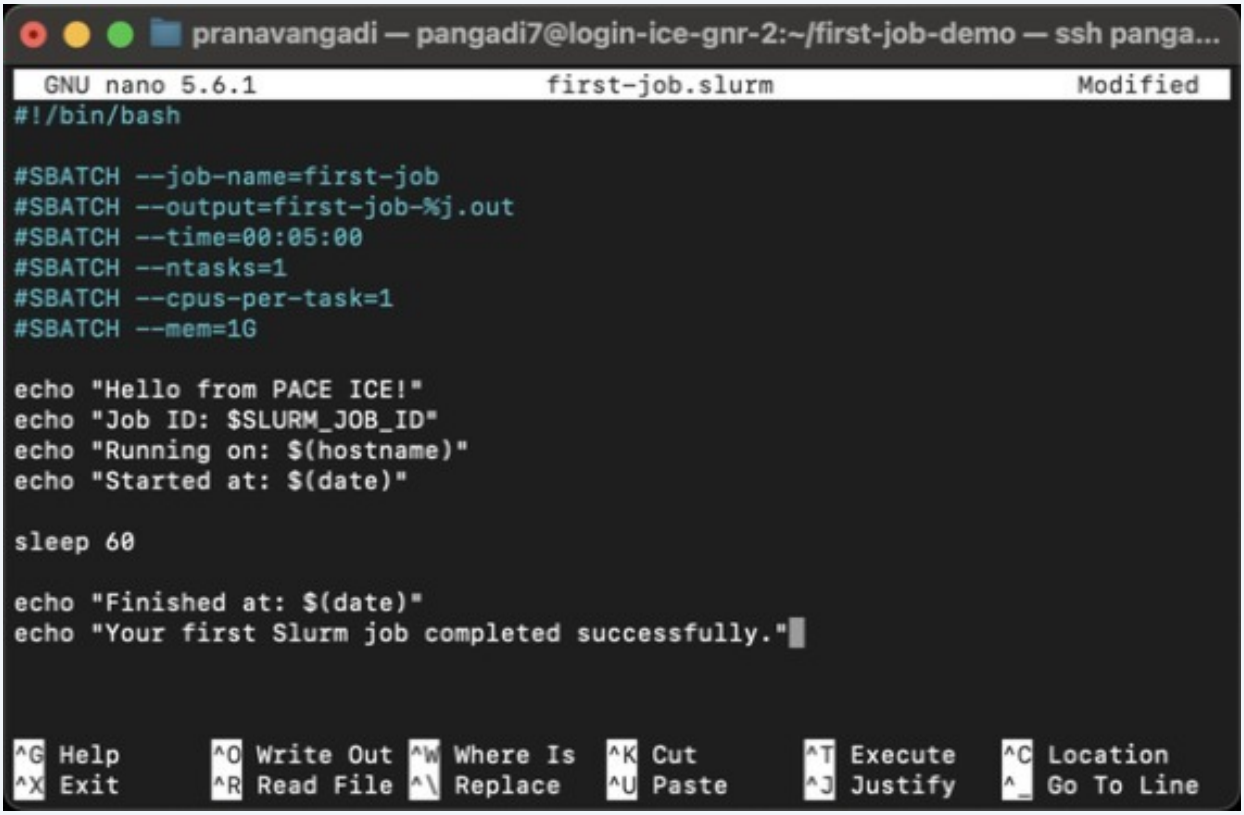


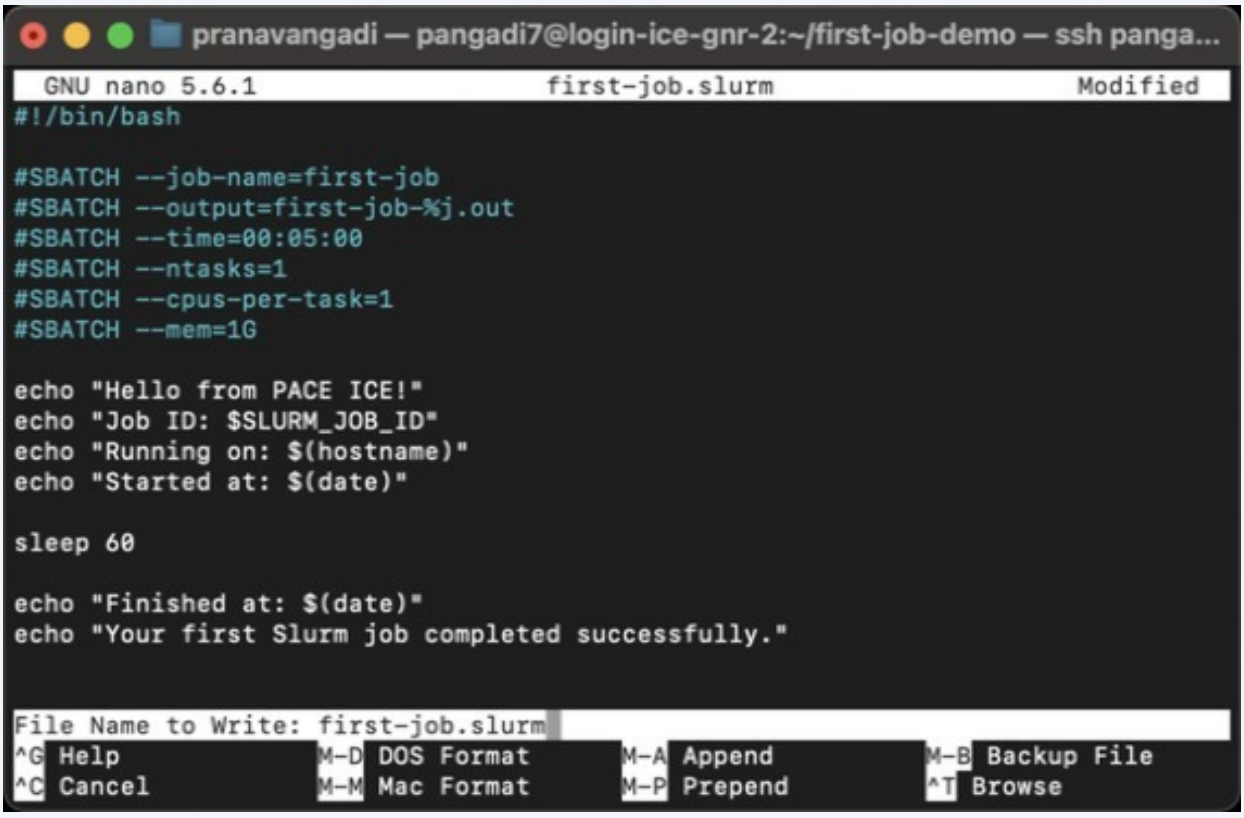
Create the Job Script

4 Open a new Slurm script in Nano
Type "**nano first-job.slurm Return**"
It will open a window as shown below.

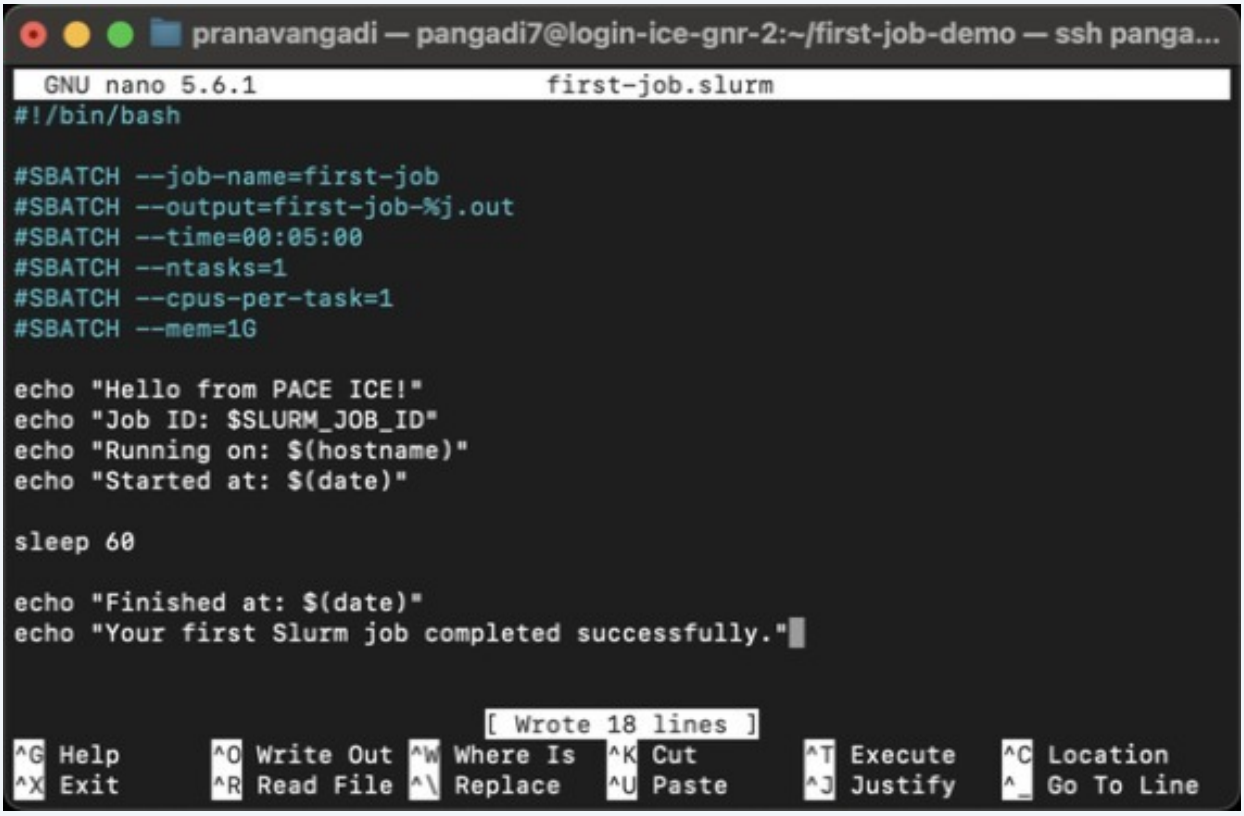


5 Paste and Save the Slurm script as first-job.slurm
Paste script then Type "**Ctrl** + **O Enter** "


pranavangadi — pangadi7@login-ice-gnr-2:~/first-job-demo — ssh panga...
GNU nano 5.6.1 first-job.slurm Modified
#!/bin/bash
#SBATCH --job-name=first-job
#SBATCH --output=first-job-%j.out
#SBATCH --time=00:05:00
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=1
#SBATCH --mem=1G
echo "Hello from PACE ICE!"
echo "Job ID: $SLURM_JOB_ID"
echo "Running on: $(hostname)"
echo "Started at: $(date)"
sleep 60
echo "Finished at: $(date)"
echo "Your first Slurm job completed successfully."
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^_ Go To Line


pranavangadi — pangadi7@login-ice-gnr-2:~/first-job-demo — ssh panga...
GNU nano 5.6.1 first-job.slurm Modified
#!/bin/bash
#SBATCH --job-name=first-job
#SBATCH --output=first-job-%j.out
#SBATCH --time=00:05:00
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=1
#SBATCH --mem=1G
echo "Hello from PACE ICE!"
echo "Job ID: $SLURM_JOB_ID"
echo "Running on: $(hostname)"
echo "Started at: $(date)"
sleep 60
echo "Finished at: $(date)"
echo "Your first Slurm job completed successfully."
File Name to Write: first-job.slurm
^G Help M-D DOS Format M-A Append M-B Backup File
^C Cancel M-M Mac Format M-P Prepend ^T Browse

6 Exit Nano After Saving
Press **Ctrl** + **X** to exit



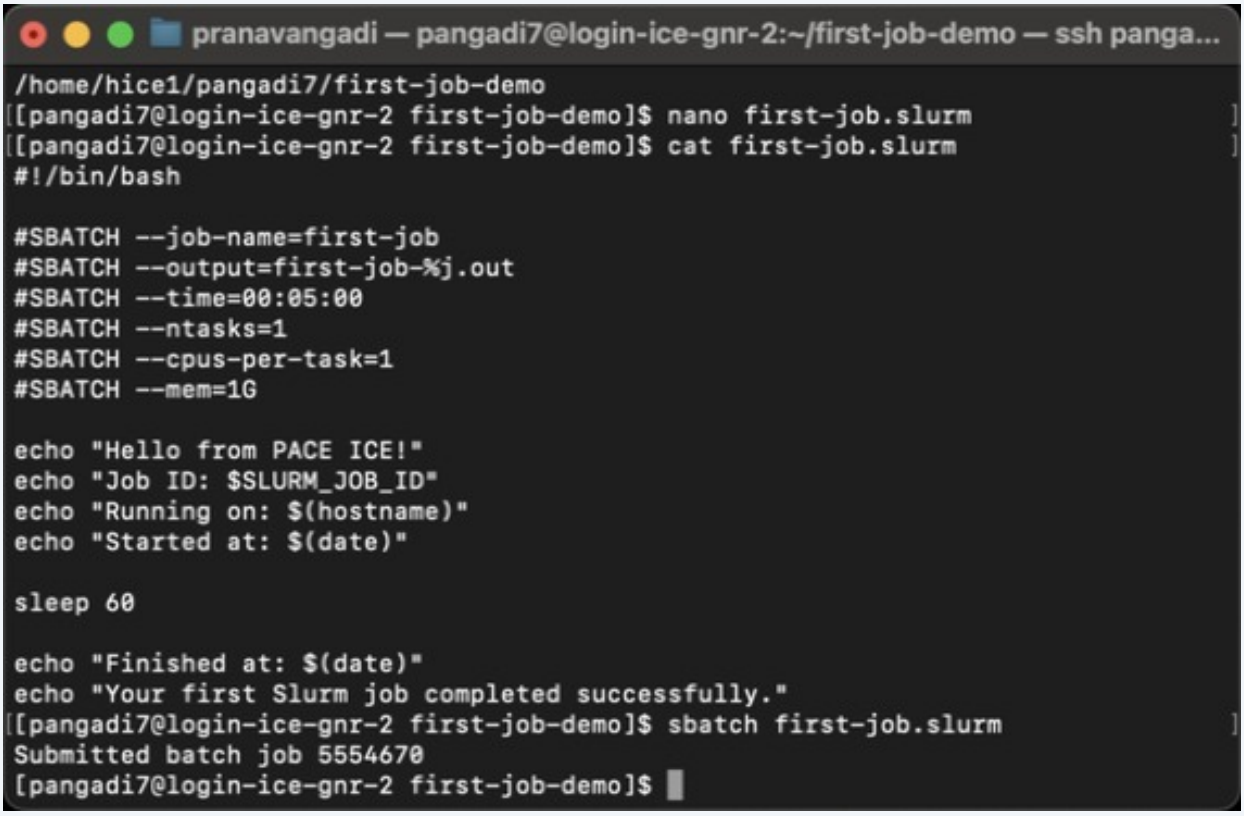
Tip!

Display the file to make sure it was saved correctly:

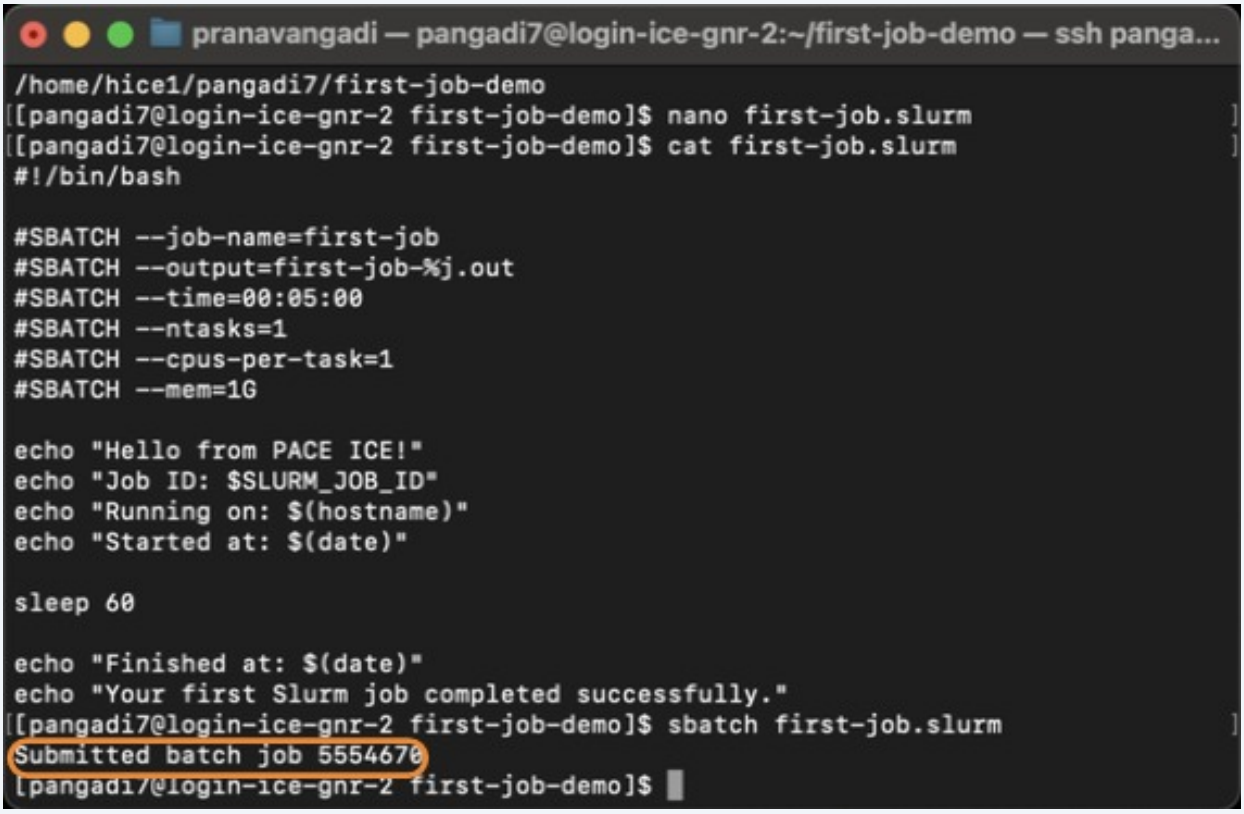
cat first-job.slurm

Submit the Job

7 Submit the job to Slurm
Type "**sbatch first-job.slurm Return**"



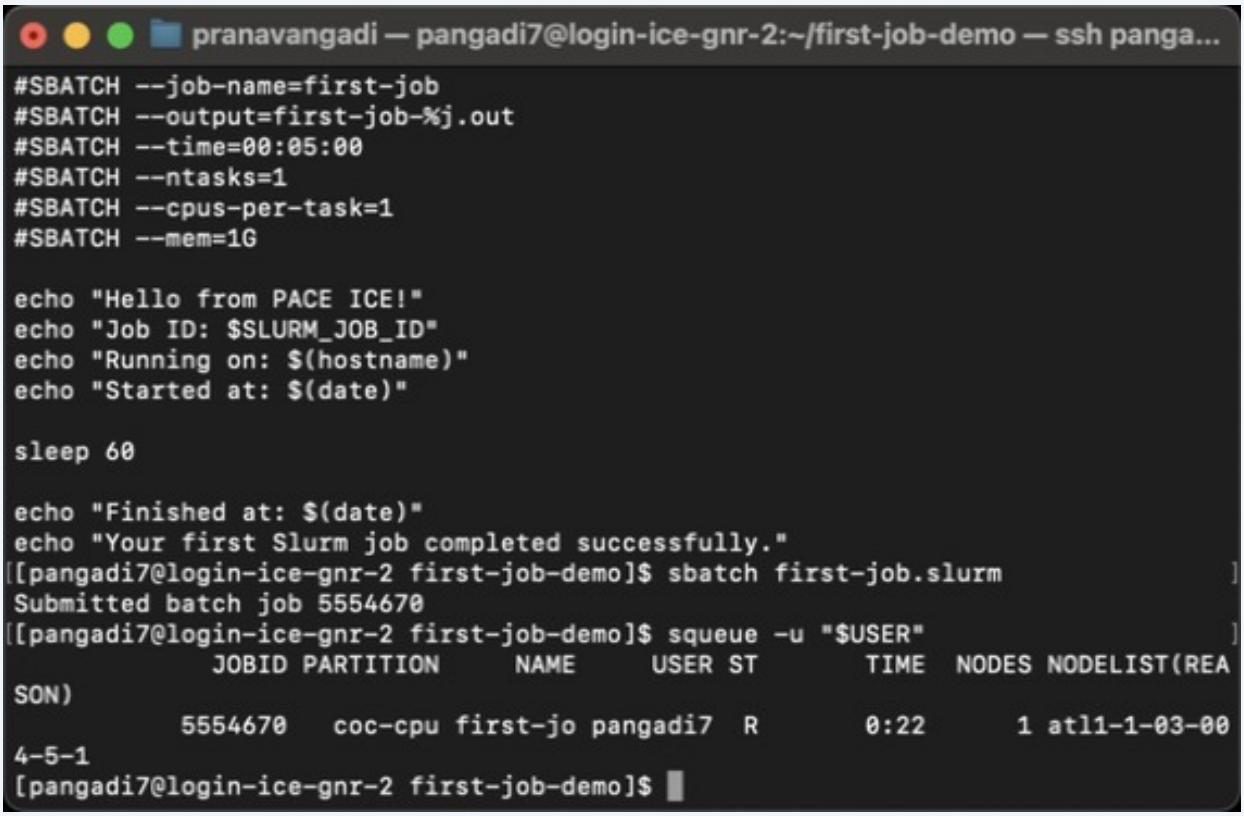
8 This number is your job ID. Your number will be different!



Monitor Job Status

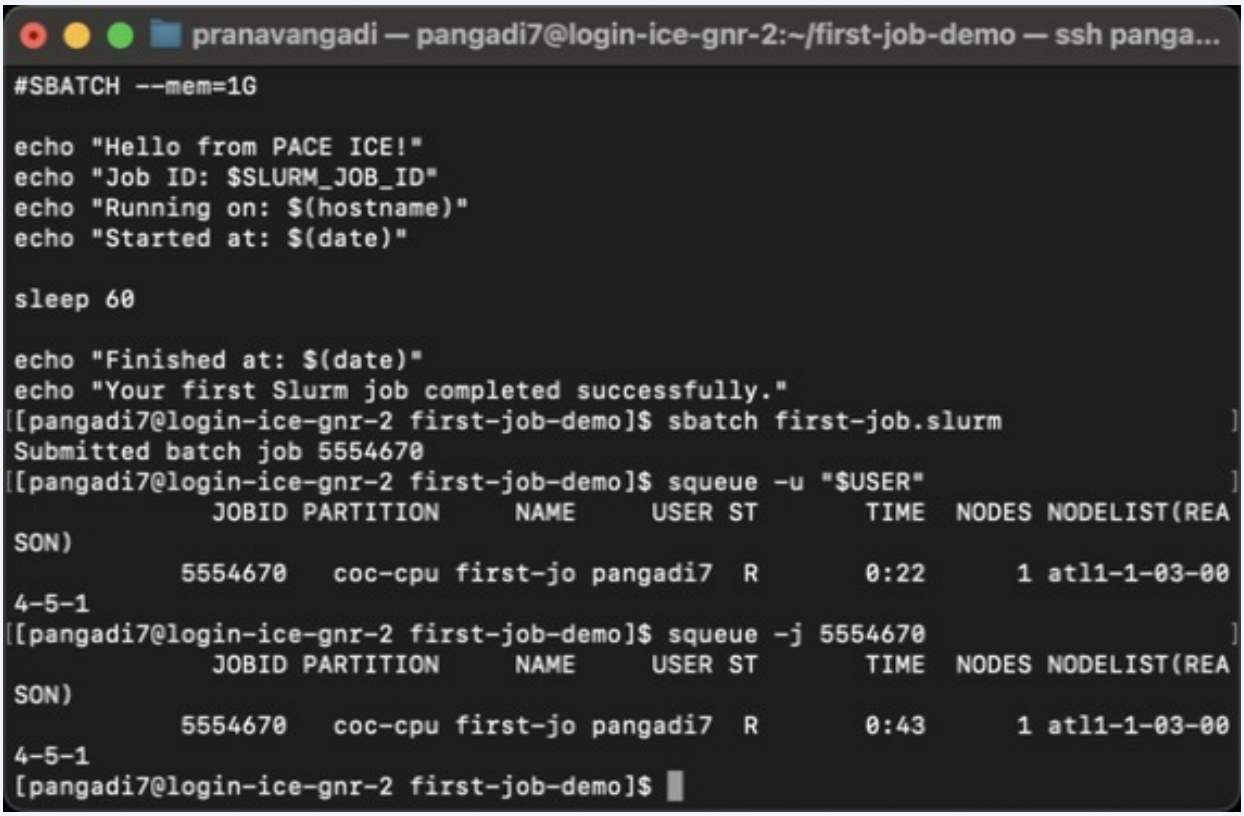
9 Check your job in the queue

Type "**squeue -u "$USER" Return**"



10 Check your job by its ID

Type "**squeue -j 5554670 Return**"

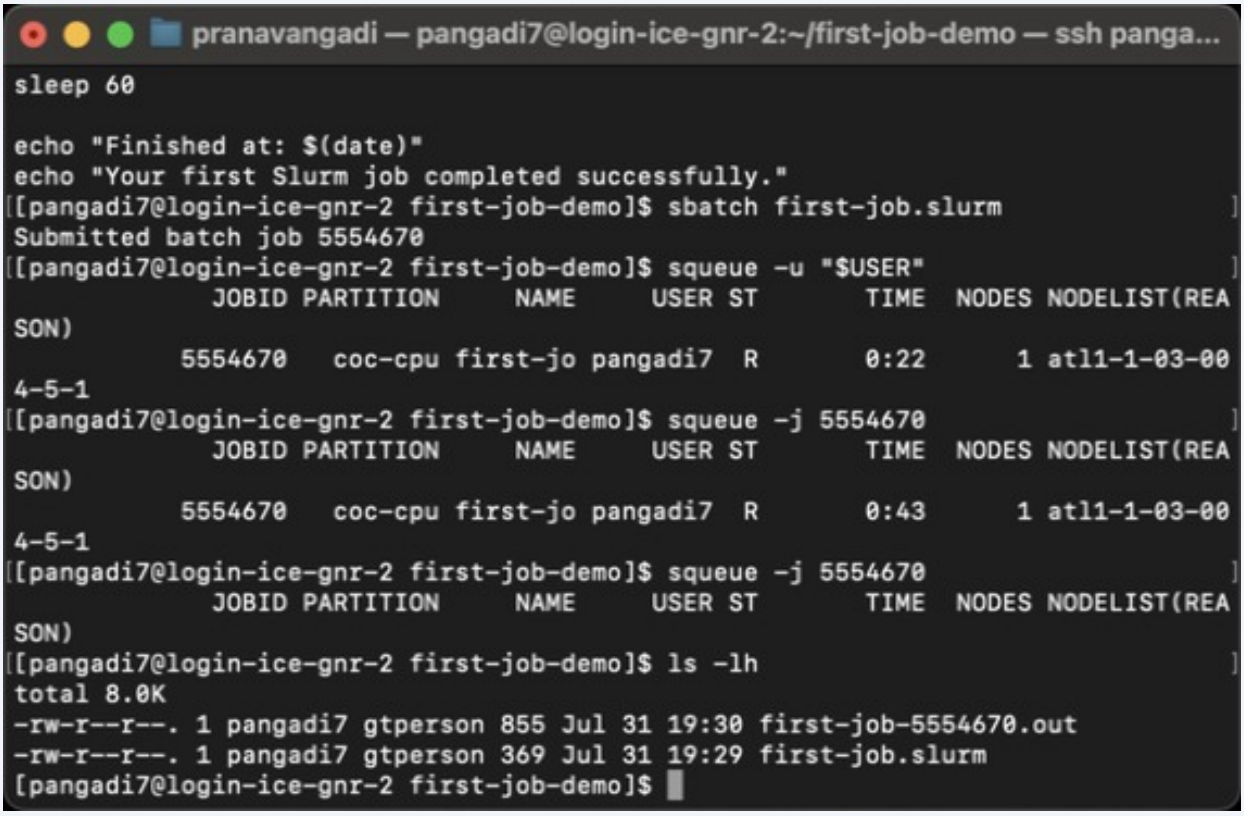


Tip!

- R means the job is running.
- PD means the job is waiting to run.
- Wait about one minute and check again. If the job no longer appears, it has left the queue.

Verify Job Output

11 Find the Slurm output file.
Type "**ls -lh Return**"



12 Read the job's output.

Type "**cat first-job-5554670.out Return**"

A successful job should end with "Your first Slurm job completed successfully"

